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ORIGINAL RESEARCH

Estimating the Price Elasticity of Cannabis Use Among U.S. Adults: Evidence from States with Recreational Cannabis Commercialization

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Abstract

Introduction: Following the spread of recreational cannabis legalization and commercialization, cannabis has become increasingly available at lower prices. As policies regulating prices are common tools to control the demand for commercialized drugs, it is crucial to understand how cannabis use responds to price changes. In this study, we assessed the association between wholesale prices for legal cannabis flower and adults' self-reported current cannabis use in ten states with recreational cannabis commercialization in the U.S.

Materials and Methods: We conducted a secondary data analysis using individual-level data on cannabis use from the longitudinal Population Assessment of Tobacco and Health Study, during 2015 and 2021. Our analysis included 19,812 U.S. adults from ten states that legalized recreational cannabis sales during the study period. We first conducted logistic regressions to estimate the association between state-level cannabis prices and individual current cannabis use. To address potential endogeneity of cannabis prices, we then employed generalized method of moment (GMM) estimator, using cannabis taxes as an instrumental variable (IV).

Results: IV-based GMM regressions suggested that cannabis taxes were a significant predictor of cannabis prices. However, the association between legal cannabis flower prices and adults' current cannabis use was negative but statistically insignificant (coefficient = −0.18, $p = 0.086$). Price elasticity estimates for current cannabis use ranged from −0.66 to −0.59 across different model specifications.

Conclusion: In the initial years of recreational cannabis commercialization in the U.S., the price elasticity of cannabis use among adults was negative but statistically insignificant. Given the rapid progression of commercialization, further research utilizing longer-term data is needed.

Keywords: Recreational cannabis; Recreational cannabis commercialization; Price elasticity; Cannabis price; Cannabis tax; Instrumental variable

Introduction

By the end of 2023, 24 states and Washington, D.C. legalized recreational cannabis. Twenty states took further steps to commercialize cannabis by establishing legal retail markets (recreational cannabis commercialization [RCC]). The U.S. cannabis market has grown significantly in recent years, with RCC states collectively generating $3.77 billion in tax revenue from recreational cannabis sales alone in 2022.[1] Being one of the fastest-growing industries, the U.S. legal cannabis market is expanding at a compound annual growth rate of 26%, and it is projected to reach $50 billion by 2026.[2]

As the market thrives, cannabis has become increasingly available at lower prices.[3,4] Take Oregon, a state with a long history of cannabis retail, as an example. The median prices for cannabis concentrates dropped from $45.00 per gram in March 2017 to

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$16.00 per gram in March 2024, decreasing almost two-thirds in seven years.[5] To regulate consumption and mitigate the risks of excessive use, particularly as prices decrease, price regulations, as suggested by research on tobacco and alcohol,[6–8] could be an effective policy tool. As a fundamental component of price instruments, taxation can be used to increase prices and influence purchasing behavior. Clear evidence indicated that imposing taxes, such as excise taxes, value added taxes, and general sales taxes, on tobacco and alcohol products were significantly associated with decreased consumptions, discouraged initiations, and promoted cessations.[9–11] Similarly, increasing cannabis prices through the implementation of excise taxes[12] is likely to decrease affordability and, consequently, reduce overall consumption.

Despite previous studies on cannabis price elasticity,[13,14] there is limited research specifically examining the price elasticity of cannabis use among U.S. adults. Moreover, most of these studies rely on price data from illicit cannabis markets existed prior to the commercialization of recreational cannabis, and the findings are mixed. For instance, one study involving 926 students at the University of California Los Angeles estimated the price elasticity of cannabis demand to range from −1.51 to −0.40.[15] Studies focused on college students across U.S. found price elasticity estimates for monthly and annual cannabis use at −0.24[16] and −0.19,[17] respectively. One nationwide study estimated the average price elasticity of cannabis demand at −0.44 across all age groups.[18] One study conducted in the 48 contiguous states found price elasticity of cannabis demand ranges from −0.79 to −0.67.[19] One study conducted nationwide reported an insignificant association between cannabis price and demand.[20] Additionally, some studies employed experimental approach to investigate cannabis price elasticities.[21,22]

When conducting price elasticity analysis, one major challenge is to obtain reliable and representative price data. Prior studies often used self-reported price data from crowdsourcing platforms[19] (e.g., priceofweed.com), local surveys,[15] or enforcement records on acquisitions of illicit cannabis prices[18,20] (e.g., the system to retrieve information from drug evidence). Since these price data were self-reported and often not market representative, the estimated effects of price might be biased, limiting their implications. Additionally, several studies employed cross-sectional designs, which overlooked time trends,[15,19] indicating associations but not causality.

To address these research gaps, we conducted this study to quantify the price elasticity of current cannabis use in legal markets. Based on the economic law of demand, we hypothesized that cannabis use among adults is negatively correlated with the price of legal cannabis. Our study makes several important contributions to the literature. First, to the best of our knowledge, it is the first study to assess cannabis price elasticity in states with legalized recreational cannabis sales, offering new empirical insights in the context of ongoing cannabis liberalization. Second, the price data used in this analysis are reliable and representative at the state level, ensuring a robust assessment of price elasticity. Third, to address potential endogeneity of cannabis prices, we use standardized cannabis taxes as an instrumental variable (IV). Finally, by utilizing a repeated cross-sectional design, our study captures variations in adult cannabis use in relation to changing cannabis prices and taxes over time.

Materials and Methods

This individual-based, secondary data analysis utilized three core datasets, including (1) individual data on past 30-day cannabis use from the Population Assessment of Tobacco and Health (PATH) Study, restricted-use files; (2) state-level legal cannabis wholesale price data from cannabis benchmarks; and (3) standardized state-level excise taxes on recreational cannabis. Our study focused on states that commercialized recreational cannabis. Ten states that had information in all three core datasets during 2015–2017 and 2019–2021 and were included in the study (see Supplementary Table S1 for details on data availability). Data access and results disclosure were reviewed and approved by Inter-university Consortium for Political and Social Research at Michigan University. All data were de-identified.

Data

The PATH study is an ongoing, nationally representative longitudinal cohort study of U.S. civilians aged 12 and above, administered by Westat and approved by its institutional review board. Recruitment for the initial Wave was conducted using a stratified address-based, area-probability sampling design during 2013–2014 and the subsequent Waves took place annually. In Wave 4, the sample was replenished with a probability supplement to address sample attrition. For the purpose of this study, we used data for adults recruited

during Wave 1–6, as well as a special collection of PATH-Adult Telephone Survey data.

Cannabis wholesale price data were sourced from cannabis benchmarks,[23] which collects prices primarily from three sources, including cultivators and dispensaries throughout the U.S.; licensed market participants within their price contributor network; and collaborated exchanges, marketplaces, brokers, vendors, and associations. Cannabis benchmarks tracks legal wholesale prices in 19 state markets that have legalized medical and/or recreational sales, releasing transaction data for each state on a weekly basis. Currently, these data only include wholesale prices for cannabis flowers, unadjusted for potency. To validate the representativeness of these price data, we compared them with official state-released price data, and found cannabis benchmarks' prices to be consistent with the official figures. Unlike prior studies that used price data from illicit cannabis markets, this study exclusively used legal cannabis price data, ensuring greater reliability and representativeness. Since PATH study was conducted across varied time spans in different Waves, we calculated average annual cannabis prices for each corresponding PATH wave period.

Given that cannabis excise taxes can vary widely in structure, ranging from ad valorem taxes (a percentage of wholesale or retail prices) to weight-based taxes on flowers or flower-derived edibles, and even potency-based taxes tied to the level of tetrahydrocannabinol and can be applied at different stages of the supply chain (e.g., cultivation, wholesale, and retail), it is crucial to standardize tax measures so as to facilitate comparability across states (see Supplementary Table S2 for details on excise tax policies). In this study, we standardized these varying tax structures by converting them into a common per-pound tax rate for cannabis flower, ensuring consistency across states and years[24] (see Supplementary Table S3 for the standardized cannabis tax rates).

Considering the potential impact of tobacco control policies on cannabis use, this study also controlled for state-level cigarette and e-cigarette taxes. Specifically, we extracted cigarette tax data from the State Tobacco Activities Tracking and Evaluation System of the Centers for Disease Control and Prevention. We used state-level e-cigarette taxes standardized by Cotti et al.[25]

Since cannabis price data have been available since 2015, after merging all datasets, our study was conducted among ten RCC states from 2015 to 2021. We excluded the year 2018 from this study due to its exclusive focus on youth surveys. The final analytic sample included 19,812 adults aged 21 and older.

Measures

The primary outcome variable in this study was current cannabis use, defined by a binary indicator that equals one if a respondent reported cannabis use during the past 30 days and zero otherwise. The main predictor of interest was the standardized cannabis wholesale price, in terms of dollars per pound, which was log-transformed (see Supplementary Fig. S1 for trends of cannabis wholesale prices by states). Additional predictors included standardized taxes on cannabis (dollars per pound), cigarettes (dollars per pack of 20 cigarettes), and e-cigarettes (dollars per fluid milliliter). Sociodemographic covariates included individual sex at birth, age, race/ethnicity, educational attainments, income levels, and marital status.

Statistical analysis

We presented summary statistics for the analytic sample. To assess the effects of price on adults' current cannabis use, we initially fitted logistic regressions, utilizing cannabis use status as the outcome variable and standardized cannabis prices as the main predictor. Considering taxation is a crucial price-based instrument, and one study indicating that cannabis taxes were largely passed through to consumers in cannabis marketplace,[12] we subsequently performed the reduced-form regression, with cannabis use as the outcome and standardized cannabis tax as the main predictor. Due to the endogeneity of cannabis prices, we further applied IV-based generalized method of moment (GMM) estimators, using cannabis tax as an IV for price.[26] In the first stage, we regressed cannabis prices on cannabis taxes. In the second stage, we regressed adult cannabis use on predicted values of prices from the first stage. To test the robustness of our results, the following covariates were sequentially added to regressions: individual demographics, cigarette taxes, and e-cigarette taxes. All regressions were conducted with standard two-way fixed effects to account for state- and time-invariant heterogeneities. Standard errors were clustered at the state level. All protocols were approved by the University of California San Diego Institutional Review Board. All statistical analyses were conducted using Stata 18.

Results

Table 1 reports summary statistics for the study sample. Approximately one-third of the respondents were current cannabis users. The average wholesale price for legal cannabis was about $1,939.28 per pound, and the average cannabis excise tax was about $714.94 per pound. The average cigarette tax was $2.44 per pack, and the average e-cigarette tax was $0.38 per fluid milliliter.

Table 2 presents associations between adult current cannabis use and legal cannabis prices by logistic regressions. Across various specifications, the results consistently suggest a negative although statistically insignificant association between legal cannabis prices and current cannabis use by adults. The estimated price elasticity ranged from −0.12 to −0.10, depending on particular function form. In addition, the findings indicated that current cannabis use was lower among individuals with higher income levels and those were married or cohabitating.

Table 3 shows the estimated associations between adult current cannabis use and standardized cannabis excise taxes. No evidence was found to support the association (coefficient = −0.000064, $p = 0.089$).

Table 4 outputs estimates through GMM instrumenting cannabis tax. In the first-stage regression, tax was validated as a significant predictor of price (coefficient = 0.00036, $p = 0.001$). Robust F-statistics (F-statistic = 2257.12) in the reduced-form equations further supported the strength of tax as an instrument, confirming the suitability of the IV-based GMM approach. Consistent with prior findings, in the second-stage regression, the association between cannabis prices and adult current cannabis use remained statistically insignificant. Accounting for all relevant predictors, as shown in the last column, 1% increase in the cannabis price corresponds to 0.18 percentage points decrease in adult cannabis use. Given that the average adult cannabis prevalence rate was around 28.84%, the reduction of 0.18 percentage point equates to approximately 0.62% of the current cannabis use. Alternatively, the estimated price elasticity for adult cannabis use was −0.62. However, this estimate was not statistically significant.

Discussion

Price-based policy tools, such as price controls and taxation, are commonly used to regulate cannabis consumption and mitigate potential misuse. Therefore, it is crucial to understand how prices influence cannabis use. This study examined the price elasticities of current cannabis use among U.S. adults in states that

Table 1. Summary Statistics on Outcome and Covariates (n = 19,812)

	Mean (%)	95% Confidence Interval
Individual-level cannabis use		
Current cannabis use	28.84	(28.20, 29.47)
State-level cannabis price and tax		
Price adjusted for inflation 2021 (dollar/pound)	1939.28	(1629.44, 2249.13)
Tax adjusted for inflation 2021 (dollar/pound)	714.94	(615.21, 814.66)
Individual-level demographic characteristics		
Male	48.59	(47.89, 49.28)
Age	40.59	(40.36, 40.83)
Race: Hispanics	26.11	(25.50, 26.73)
Race: Non-Hispanic Whites	57.021	(56.33, 57.71)
Race: Non-Hispanic Blacks	5.93	(5.60, 6.26)
Race: Non-Hispanic Others	10.060	(9.64, 10.48)
Race: Unknown	1.11	(0.96, 1.26)
High school or below	30.60	(29.96, 31.24)
Some college	38.061	(37.38, 38.74)
College and above	31.34	(30.69, 31.99)
Household income: ≤ $24,999	25.93	(25.32, 26.54)
Household income: between $25,000 and $49,999	21.28	(20.71, 21.85)
Household income: between $50,000 and $99,999	25.91	(25.30, 26.52)
Household income: ≥ $100,000	22.032	(21.45, 22.61)
Household income: Unknown	4.85	(4.55, 5.15)
Married/cohabiting	40.48	(39.79, 41.16)
State-level tobacco tax controls		
Cigarette tax adjusted for inflation 2021 (dollar/pack of 20 cigarettes)	2.44	(2.11, 2.77)
E-cigarette tax adjusted for inflation 2021 (dollar/fluid milliliter)	0.38	(0.16, 0.60)

Notes: Statistics for cannabis price, cannabis tax, cigarette tax, and e-cigarette tax are generated at state-year level.

Table 2. Cannabis Price as Main Predictor: Logistic Regressions Without Instrumental Variable (n = 19,507)

	Model 1	Model 2	Model 3	Model 4
Price	−0.026 (−0.090, 0.037)	−0.033 (−0.11, 0.040)	−0.028 (−0.092, 0.037)	−0.036 (−0.11, 0.041)
Implied price elasticity	−0.10	−0.11	−0.10	−0.12
State-level substance covariates				
Cigarette tax	—	0.017 (−0.0036, 0.037)	—	0.017 (−0.0036, 0.038)
E-cigarette tax	—	—	0.0031 (−0.023, 0.029)	0.0070 (−0.017, 0.031)
Demographic characteristics				
Married/Cohabiting	−0.10*** (−0.13, −0.083)	−0.10*** (−0.13, −0.083)	−0.10*** (−0.13, −0.083)	−0.10*** (−0.13, −0.083)
Male	0.061*** (0.042, 0.080)	0.0.61*** (0.042, 0.080)	0.061*** (0.042, 0.080)	0.061*** (0.042, 0.080)
Hispanic	−0.095*** (−0.13, −0.060)	−0.095*** (−0.13, −0.060)	−0.095*** (−0.13, −0.060)	−0.095*** (−0.13, −0.060)
Non-Hispanic Blacks	0.0026 (−0.023, 0.028)	0.0024 (−0.023, 0.028)	0.0026 (−0.023, 0.028)	0.0024 (−0.023, 0.028)
Non-Hispanic Others	−0.042 (−0.084, 0.00074)	−0.041 (−0.084, 0.0010)	−0.042 (−0.084, 0.00073)	−0.041 (−0.084, 0.0010)
Unknown Race	−0.024 (−0.12, 0.069)	−0.025 (−0.12, 0.069)	−0.024 (−0.12, 0.069)	−0.025 (−0.12, 0.069)
Age	−0.0048*** (−0.0054, −0.0042)	−0.0048*** (−0.0054, −0.0042)	−0.0048*** (−0.0054, −0.0042)	−0.0048*** (−0.0054, −0.0042)
Some college	0.0063 (−0.015, 0.028)	0.0061 (−0.016, 0.028)	0.0063 (−0.015, 0.028)	0.0061 (−0.016, 0.028)
Bachelor	−0.020 (−0.048, 0.0073)	−0.020 (−0.048, 0.0072)	−0.020 (−0.048, 0.0073)	−0.020 (−0.048, 0.0072)
Income 25k–50k	−0.035* (−0.068, −0.0020)	−0.035* (−0.068, −0.0020)	−0.035* (−0.068, −0.0020)	−0.035* (−0.068, −0.0021)
Income 50k–100k	−0.072*** (−0.11, −0.037)	−0.072*** (−0.11, −0.037)	−0.072*** (−0.11, −0.037)	−0.072*** (−0.11, −0.037)
Income 100k above	−0.10*** (−0.16, −0.041)	−0.10*** (−0.16, −0.041)	−0.10*** (−0.16, −0.041)	−0.10*** (−0.16, −0.041)
Unknown income	−0.11*** (−0.16, −0.064)	−0.11*** (−0.16, −0.064)	−0.11*** (−0.16, −0.064)	−0.11*** (−0.16, −0.064)

Marginal effects and 95% CIs are reported. All models are conducted with logistic regressions, and controlled for state and year fixed effects. Price elasticity was estimated by dividing the coefficient estimate by the prevalence of current cannabis use and then multiplying the result by 100%.

*$p < 0.05$.

***$p < 0.001$.

legalized recreational cannabis sales. Through various model specifications, the findings indicated that during the initial years following recreational cannabis commercialization, current cannabis use by adults was not significantly responsive to cannabis price changes.

The implied elasticity of cannabis use with respect to cannabis prices was approximately −0.62, suggesting a relatively inelastic demand for cannabis. This estimate is consistent with previous research, which reported price elasticities ranging from −0.79 to −0.67, using data from both legal and illicit cannabis markets.[19] Given that cannabis prices decrease following the commercialization,[4] and considering that price elasticity is generally higher for more expensive goods, it is reasonable that our estimated price elasticity is somewhat lower. While other studies have also identified inelastic demand for cannabis use in certain specifications,[15–18,20] the magnitude of the effect may vary due to differences in data types, the cannabis legalization landscapes and the analytic approaches.

In this study, we found insignificant association between legal cannabis price and current cannabis use. There are several explanations for this insignificant estimate. First, our price measure reflects legal cannabis prices, but adult cannabis users may obtain cannabis from both legal and illicit markets. Due to the perception that legal cannabis products tend to be more expensive than illicit ones,[27] despite a transition to a legal marketplace, users may still opt for vendors with established relationships, such as unlicensed dispensaries or illicit online sellers. Consequently, this could lead to a statistically insignificant association between legal prices and use status. Second, it takes several years to establish a legal marketplace following legalization, and this process might be further delayed due to enforcement across states.[28] In our study, most of the states had only legalized recreational cannabis sales for two or three years. The scarcity of data from legal sales era limits our statistical power to detect significant price effects, which explain why we estimated negative but nonsignificant effects of prices on use. Third, as suggested by previous research,[19] the demand for cannabis could be inelastic at the extensive margin (i.e., use status). Compared to cannabis consumption and sales (i.e., intensive margin of consumption), cannabis use status may be less sensitive to

Table 3. Cannabis Tax as Main Predictor: Logistic Regressions without Instrumental Variable (n = 19,507)

	Model 1	Model 2	Model 3	Model 4
Cannabis tax	−0.000061 (−0.00013, 0.000012)	−0.000064 (−0.00014, 9.06e-06)	−0.000062 (−0.00013, 0.000012)	−0.000064 (−0.00014, 9.73e-06)
Implied tax elasticity	−0.00021	−0.00022	−0.00021	−0.00022
State-level substance covariates				
Cigarette tax	—	0.017 (−0.0024, 0.036)	—	0.017 (−0.0025, 0.036)
E-cigarette tax	—	—	−0.0019 (−0.035, 0.031)	0.0012 (−0.032, 0.034)
Demographic characteristics				
Married/Cohabiting	−0.10*** (−0.12, −0.091)	−0.10*** (−0.12, −0.091)	−0.10*** (−0.12, −0.091)	−0.10*** (−0.12, −0.091)
Male	0.061*** (0.049, 0.073)	0.061*** (0.049, 0.073)	0.061*** (0.049, 0.073)	0.061*** (0.049, 0.073)
Hispanic	−0.095*** (−0.11, −0.078)	−0.095*** (−0.11, −0.078)	−0.095*** (−0.11, −0.078)	−0.095*** (−0.11, −0.078)
Non-Hispanic Blacks	0.0024 (−0.023, 0.028)	0.0022 (0.024, 0.028)	0.0024 (−0.023, 0.028)	0.0022 (−0.024, 0.028)
Non-Hispanic Others	−0.041*** (−0.063, −0.020)	−0.041*** (−0.062, −0.020)	−0.041*** (−0.063, 0.020)	−0.041*** (−0.062, −0.020)
Unknown Race	−0.023 (−0.17, 0.12)	−0.024 (−0.17, 0.12)	−0.023 (−0.17, 0.12)	−0.024 (−0.17, 0.12)
Age	−0.0048*** (−0.0052, −0.0044)	−0.0048*** (−0.0052, −0.0044)	−0.0048*** (−0.0052, −0.0044)	−0.0048*** (−0.0052, −0.0044)
Some college	0.0063 (−0.0089, 0.021)	0.0061 (−0.0091, 0.021)	0.0063 (−0.0089, 0.021)	0.0061 (−0.0091, 0.021)
Bachelor	−0.020* (−0.039, −0.0021)	−0.020* (−0.039, −0.0022)	−0.020* (−0.039, −0.0021)	−0.020* (−0.039, −0.0022)
Income 25k–50k	−0.035* (−0.052, −0.017)	−0.035*** (−0.052, −0.017)	−0.035* (−0.052, −0.017)	−0.035*** (−0.052, −0.017)
Income 50k–100k	−0.072*** (−0.090, −0.054)	−0.072*** (−0.090, −0.054)	−0.072*** (−0.090, −0.054)	−0.072*** (−0.090, −0.054)
Income 100k above	−0.10*** (−0.12, −0.080)	−0.10*** (−0.12, −0.080)	−0.10*** (−0.12, −0.080)	−0.10*** (−0.12, −0.080)
Unknown income	−0.11*** (−0.14, −0.078)	−0.11*** (0.14, −0.078)	−0.11*** (−0.14, −0.078)	−0.11*** (−0.14, −0.078)

Marginal effects and 95% CIs are reported. All models are conducted with logistic regressions, and controlled for state and year fixed effects. When estimating implied tax elasticity, we assume cannabis excise taxes are fully passed through to prices. Tax elasticity was estimated by dividing the coefficient estimate by the prevalence of current cannabis use and then multiplying the result by 100%.

*$p < 0.05$.

***$p < 0.001$.

prices. In addition, as demonstrated in the rational addiction model proposed by Becker and Murphy,[29] long-term price responses are generally more sensitive than short-run responses for addictive goods. Therefore, it may take time for cannabis use status to respond to prices. Future research is encouraged to expand the legal price elasticity analysis over a longer term. Since prices did not seem to influence cannabis use during the initial years of recreational cannabis commercialization, policies regulating alternative demand and supply factors could be considered. These may include limiting the number of cannabis dispensaries, reducing operating hours for cannabis businesses, and implementing restrictions on cannabis advertising.

To our knowledge, this study took the first attempt to quantify price elasticity of cannabis use using legal cannabis price data. While previous studies have explored similar associations among U.S. adults,[15,16,18–20] our study focused exclusively on states with legal cannabis markets. Another strength of the study was to rely on reliable and nationally representative price data. In this regard, our study closely characterized marketplaces in recreational cannabis commercialized states through a wide presentation of legal cannabis prices. Unlike self-reported prices, which might be susceptible to reporting bias and measurement errors, our price data accurately captured the diversity and complexity of each cannabis marketplace. The high reliability and validity of the data increased the confidence in research findings and their policy implications. Compared to existing literature on price elasticity for cannabis demand by adults, this study is the first to consider the effect of cannabis tax and address the endogeneity of prices by using cannabis taxes as an IV.

This study has limitations. First, since research on recreational cannabis commercialization is just emerging, this study was limited to only ten states over a relatively short timeframe. This resulted in minimal variations in prices both across states and over time. Second, the cannabis market offers a diverse range of products, including flowers, edibles, tinctures, concentrates, and more. The price elasticity of cannabis use may vary significantly by product types. In this study, we only focused on flower prices, the best-selling cannabis product, which constitutes more than half of the market share.[30] Future research is encouraged to provide insightful analysis by differentiating product type. Third, although all respondents lived in states where recreational cannabis sale is legal, it

Table 4. Past-Month Cannabis Use Estimation: Generalized Method of Moments Regressions with Cannabis Tax as Instrumental Variable (n = 19,507)

	Model 1	Model 2	Model 3	Model 4
First Stage				
Cannabis tax	0.00034*** (0.00033, 0.00036)	0.00034*** (0.00032, 0.00035)	0.00036*** (0.00035, 0.00038)	0.00036*** (0.00034, 0.00037)
State level substance covariates				
Cigarette tax	—	0.022*** (0.018, 0.026)	—	0.027*** (0.024, 0.031)
E-cigarette tax	—	—	0.094*** (0.089, 0.099)	0.099*** (0.094, 0.10)
Demographic characteristics				
Married/Cohabiting	0.00082 (−0.0016, 0.0032)	0.00073 (−0.0017, 0.0031)	0.00074 (−0.0016, 0.0031)	0.00053 (−0.0017, 0.0029)
Male	−0.00029 (−0.0024, 0.0018)	−0.00019 (−0.0023, 0.0019)	−0.00039 (−0.0024, 0.0016)	−0.00027 (−0.0023, 0.0017)
Hispanic	−0.00029 (−0.0031, 0.0025)	−0.00052 (−0.0033, 0.0023)	−0.00012 (−0.0029, 0.0026)	−0.00039 (−0.0031, 0.0023)
Non-Hispanic Blacks	0.0036* (0.00026, 0.0074)	0.0037* (0.00010, 0.0073)	0.0034 (−0.00010, 0.0068)	0.0031 (−0.00031, 0.0066)
Non-Hispanic Others	−0.00056 (−0.0042, 0.0031)	−0.00036 (−0.0040, 0.0033)	−0.00041 (−0.0040, 0.0031)	−0.00017 (−0.0037, 0.0034)
Unknown race	−0.0034 (−0.023, 0.016)	−0.0043 (−0.024, 0.015)	−0.0031 (−0.022, 0.016)	−0.0042 (−0.023, 0.015)
Age	−0.000020 (−0.00010, 0.000047)	−0.000020 (−0.00010, 0.000047)	−0.000015 (−0.00010, 0.000051)	−0.000015 (−0.00010, 0.000051)
Some college	−0.0014 (−0.0041, 0.0013)	−0.0016 (−0.0043, 0.0010)	−0.0014 (−0.0040, 0.0012)	−0.0016 (−0.0042, 0.0010)
Bachelor	−0.00010 (−0.0032, 0.0031)	−0.00026 (−0.0034, 0.0029)	−0.00044 (−0.0035, 0.0026)	−0.00069 (−0.0037, 0.0023)
Income 25k–50k	−0.00045 (−0.0037, 0.0028)	−0.00032 (−0.0035, 0.0029)	−0.00039 (−0.0035, 0.0027)	−0.00023 (−0.0033, 0.0029)
Income 50k–100k	−0.0010 (−0.0042, 0.0022)	−0.0011 (−0.0043, 0.0021)	−0.0010 (−0.0036, 0.0026)	−0.0010 (−0.0040, 0.0025)
Income 100k above	0.00054 (−0.0030, 0.0040)	0.00073 (−0.0028, 0.0042)	0.00068 (−0.0027, 0.0041)	0.00093 (−0.0025, 0.0043)
Unknown income	−0.0023 (−0.0076, 0.0029)	−0.0021 (−0.0073, 0.0031)	−0.0021 (−0.0071, 0.0030)	−0.0018 (−0.0068, 0.0032)
Weak instrument test				
F-statistic	2005.80	2067.43	2144.95	2257.12
Second Stage				
Price	−0.18 (−0.40, 0.034)	−0.19 (−0.41, 0.025)	−0.17 (−0.37, 0.033)	−0.18 (−0.39, 0.026)
Implied price elasticity	−0.62	−0.66	−0.59	−0.62
State-level substance covariates				
Cigarette tax	—	0.022* (0.0015, 0.043)	—	0.023* (0.0021, 0.044)
E-cigarette tax	—	—	0.016 (−0.021, 0.053)	0.021 (−0.017, 0.059)
Demographic characteristics				
Married/Cohabiting	−0.10*** (−0.12, −0.090)	−0.10*** (−0.12, −0.090)	−0.10*** (−0.12, −0.090)	−0.10*** (−0.12, −0.090)
Male	0.063*** (0.050, 0.075)	0.063*** (0.050, 0.075)	0.063*** (0.050, 0.075)	0.063*** (0.050, 0.075)
Hispanic	−0.098*** (−0.12, −0.080)	−0.098*** (−0.12, −0.080)	−0.098*** (−0.12, −0.080)	−0.098*** (−0.12, −0.080)
Non-Hispanic Blacks	0.0055 (−0.024, 0.035)	0.0054 (−0.024, 0.034)	0.0054 (−0.024, 0.034)	0.0052 (−0.024, 0.034)
Non-Hispanic Others	−0.042*** (−0.063, −0.021)	−0.042*** (−0.063, −0.021)	−0.042*** (−0.063, −0.021)	−0.042*** (−0.063, −0.021)
Unknown race	−0.018 (−0.15, 0.11)	−0.018 (−0.15, 0.11)	−0.018 (−0.15, 0.11)	−0.019 (−0.15, 0.11)
Age	−0.0047*** (−0.0050, −0.0043)	−0.0047*** (−0.0050, −0.0043)	−0.0047*** (−0.0050, −0.0043)	−0.0047*** (−0.0050, −0.0043)
Some college	0.0065 (−0.0095, 0.023)	0.0063 (−0.0097, 0.022)	0.0065 (−0.0095, 0.023)	0.0063 (−0.0097, 0.022)
Bachelor	−0.019* (−0.037, −0.0010)	−0.019* (−0.038, −0.0011)	−0.019* (−0.037, −0.0010)	−0.019* (−0.038, −0.0012)
Income 25k–50k	−0.039*** (−0.058, −0.020)	−0.039*** (−0.058, −0.020)	−0.039*** (−0.058, −0.020)	−0.039*** (−0.058, −0.020)
Income 50k–100k	−0.077*** (−0.096, −0.058)	−0.077*** (−0.096, −0.058)	−0.077*** (−0.096, −0.058)	−0.077*** (−0.096, −0.058)
Income 100k above	−0.10*** (−0.12, −0.082)	−0.10*** (−0.12, −0.082)	−0.10*** (−0.12, −0.082)	−0.10*** (−0.12, −0.082)
Unknown income	−0.10*** (−0.13, −0.076)	−0.10*** (−0.13, −0.075)	−0.10*** (−0.13, −0.076)	−0.10*** (−0.13, −0.076)

Same covariates are used in first- and second-stage estimations. All models are conducted with GMM regressions, using cannabis tax as IV. All models control for state and year fixed effects. Price elasticity was estimated by dividing the coefficient estimate by the prevalence of current cannabis use and then multiplying the result by 100%.

*$p < 0.05$.

***$p < 0.001$.

remains unknown whether respondents obtained cannabis from legal or illicit sources. Fourth, while cannabis prices considered in this study reflect only wholesale cost of flowers, products consumed by respondents may include a variety of cannabis categories, not just flowers. Fifth, due to the challenges in acquiring retail cannabis price data, we relied on wholesale prices instead. Despite wholesale prices exhibit similar trend as retail prices, considering in-store promotional sales and the staggered price transmission at the wholesale stage, consumers were generally more responsive to changes in retail prices. Sixth, this study estimated the price elasticity of cannabis use along the extensive margin rather than the intensive margin. It will bring improved sensitivity and precision, in future studies, to exploit the price impact on intensive margin elasticity. Seventh, the study period included the COVID-19 pandemic, which may have influenced respondents' cannabis use behaviors, particularly during stay-at-home orders. Eighth, this study focused solely on state-level policies regarding recreational cannabis commercialization, without considering local-level regulations. Last, this study included only a few state-level controls. Given the limited scope of our analysis across ten states, including multiple state-level covariates may result in multicollinearity issues.

Conclusions

This study estimated the price elasticity of cannabis use among U.S. adults. We found no evidence of an association between the price of legal cannabis and its current use. Given that recreational cannabis commercialization is still in its early stages, future studies are recommended to explore this analysis over a longer term.

Authors' Contributions

B.H., C.S., and Y.S.: Conceptualization. B.H., Y.H., H.P., C.S., and Y.S.: Data curation and investigation. C.S. and Y.S.: Funding acquisition. B.H.: Formal analysis, software, visualization, and validation. Y.S.: Supervision. B.H.: Writing—original draft. B.H., Y.H., H.P., C.S., and Y.S.: Writing—edits and approval of the final article.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT 4.0 in order to check grammar errors and improve the overall flow of paragraphs. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Author Disclosure Statement

No competing financial interests exist.

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Supplementary Material

Supplementary Figure S1
Supplementary Table S1
Supplementary Table S2
Supplementary Table S3

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Abbreviations Used

CI = confidence interval
COVID-19 = Coronavirus Disease 2019
GMM = generalized method of moment
IV = instrumental variable
PATH = Population Assessment of Tobacco and Health
RCC = recreational cannabis commercialization
U.S. = United States